# Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering

**Course Title:** Basic Electrical and Electronic Circuit

**Course Code:** EEE 0713 1206B

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## Experiment 04

**Verification of Thevenin's Theorem for a DC Network**

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Submission Date: August 25, 2026

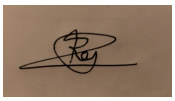
# Declaration

I hereby declare that the work presented in this laboratory report entitled **"Verification of Thevenin's Theorem for a DC Network"** is an authentic record of my own work carried out for the course **Basic Electrical and Electronic Circuit (EEE 0713 1206B)**.

This report has been prepared by me independently and has not been submitted elsewhere for the award of any degree, diploma, or credit.

Sylhet, Bangladesh

**Date:** August 25, 2026



**Raj Bhowmik**
Reg. No: 202561901048

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# Objective

- To understand the core principles of network reduction using Thevenin's theorem in linear, bilateral DC circuits.
- To construct a multi-resistor DC network and measure its open-circuit voltage ($V_{th}$) and equivalent resistance ($R_{th}$).
- To verify experimentally that the original network and its single-source Thevenin equivalent deliver identical current and voltage to a load.
- To evaluate agreement between empirical laboratory data and theoretical network calculations.

# Introduction

Analyzing load response variations in complex networks becomes computationally repetitive if the complete circuit must be re-solved for every load modification. Thevenin's Theorem addresses this by modeling any two-terminal linear network—regardless of internal complexity—as a single equivalent voltage source connected in series with an equivalent resistance. This experiment demonstrates Thevenin's theorem by assembling a DC network, determining its equivalent parameters, and demonstrating terminal equivalence across a load resistor.

# Theory

## Thevenin's Theorem

Thevenin's Theorem states that any two-terminal, linear, bilateral DC network containing voltage sources and resistors can be represented by an equivalent circuit comprising an independent voltage source ($V_{th}$) in series with an equivalent resistor ($R_{th}$).

- **Thevenin Voltage ($V_{th}$):** The open-circuit voltage measured across terminal pair A–B when load $R_L$ is removed.
- **Thevenin Resistance ($R_{th}$):** The equivalent resistance measured looking back into terminals A–B with all independent voltage sources replaced by short circuits (0 V).

Connected to load $R_L$, the load current $I_L$ and terminal voltage $V_L$ follow:

$$I_L = \frac{V_{th}}{R_{th} + R_L}, \qquad V_L = I_L R_L.$$

## Relation to Norton's Theorem

Norton's Theorem offers a dual representation: replacing the network with an equivalent current source $I_N$ in parallel with $R_N$, where $R_N = R_{th}$ and $V_{th} = I_N R_N$.

# Apparatus and Components

# Circuit Diagram

## Original Circuit

## Thevenin Equivalent Circuit

Table 1: List of Equipment and Components Used

| Item Description | Specification / Rating | Quantity |
|---|---|---|
| DC Power Supply | 15 V regulated | 1 |
| Resistor $R_1$ | 670 Ω | 1 |
| Resistor $R_2$ | 670 Ω | 1 |
| Resistor $R_3$ | 670 Ω | 1 |
| Load Resistor $R_L$ | 970 Ω | 1 |
| Digital Multimeter (DMM) | DC Voltage / Current / Ohms | 1 |
| Solderless Breadboard | Standard Tie-Point | 1 |
| Connecting Wires | Jumper Wires | As required |

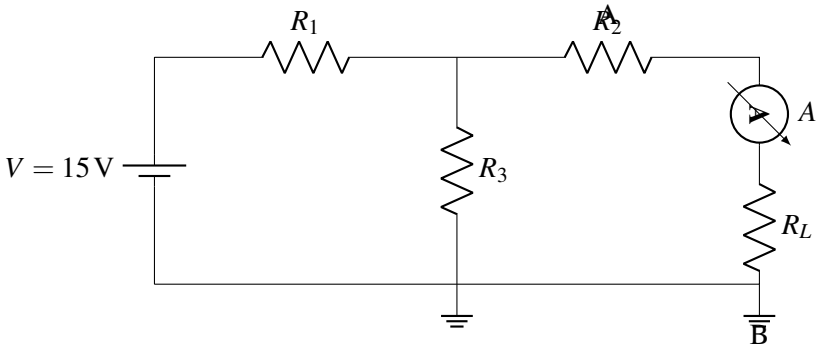



Figure 1: Original DC network featuring $R_1 = R_2 = R_3 = 670\,\Omega$ and load $R_L = 970\,\Omega$ driven by $V = 15\,\text{V}$.

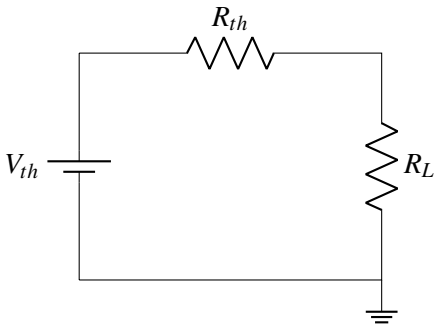



Figure 2: Thevenin equivalent circuit driving load $R_L$ via series parameters $V_{th}$ and $R_{th}$.

# Experimental Procedure

## Part A: Original Circuit Measurement

1. Assemble the primary circuit on the breadboard (Figure 1) with load $R_L$ across terminals A–B.
2. Record load voltage ($V_L$) and current ($I_L$) using the digital multimeter.

## Part B: Verification of Thevenin's Theorem

1. Disconnect load resistor $R_L$ from terminals A–B.
2. Measure open-circuit voltage across terminals A–B to determine $V_{th}$.
3. Deactivate the DC source (short-circuiting supply terminals) and measure terminal resistance across A–B using an ohmmeter to determine $R_{th}$.

4. Build the equivalent model (Figure 2) on a separate breadboard section using $V_{th}$ and $R_{th}$.
5. Connect load $R_L$ across the equivalent circuit and record resulting load parameters.

## Theoretical Calculation

With load $R_L$ removed, open-circuit output matches the voltage division across $R_3$:

$$V_{th} = V\left(\frac{R_3}{R_1 + R_3}\right).$$

With source $V$ short-circuited, equivalent input resistance equals $R_1 \parallel R_3$ in series with $R_2$:

$$R_{th} = R_2 + \left(\frac{R_1 R_3}{R_1 + R_3}\right).$$

Theoretical load parameters are computed as:

$$I_L = \frac{V_{th}}{R_{th} + R_L}, \qquad V_L = I_L R_L.$$

## Data Collection

Table 2: Experimental Data for the Verification of Thevenin's Theorem

| Parameter | Original Circuit (measured) | Thevenin Equiv. (measured) | Theoretical Value | % Error |
|---|---|---|---|---|
| Open-Circuit Voltage, $V_{th}$ | N/A | 8.23 V | 8.105 V | 1.54% |
| Equivalent Resistance, $R_{th}$ | N/A | 1.03 kΩ | 978 Ω | 5.3% |
| Load Current, $I_L$ | 4 mA | 4.02 mA | 4.16 mA | 3.37% |
| Load Voltage, $V_L$ | 4 V | 4.07 V | 4.0352 V | 0.86% |

## Result Analysis

Measured parameters ($V_{th} = 8.23\,\text{V}$, $R_{th} = 1.03\,\text{k}\Omega$) showed close agreement with theoretical values (8.105 V, 978 Ω), yielding minor percentage errors of 1.54% and 5.3%. Original load metrics ($I_L = 4\,\text{mA}$, $V_L = 4\,\text{V}$) closely matched parameters recorded from the physical equivalent circuit ($I_L = 4.02\,\text{mA}$, $V_L = 4.07\,\text{V}$), confirming electrical equivalence.

## Discussion and Sources of Error

Experimental deviations are mainly attributed to:

1. **Resistor Tolerances:** Variance between coded and real resistance values influences both $R_{th}$ measurements and theoretical outputs.
2. **Contact Resistance:** Contact resistance within breadboard strips and shorting links slightly shifts ohmmeter resistance readings.
3. **Multimeter Burden Voltage:** Ammeter internal resistance introduces minor loading effects during branch current readings.
4. **Equivalent Parameter Substitution:** Practical component tolerances prevent exact matching of calculated $R_{th}$ on physical breadboards.

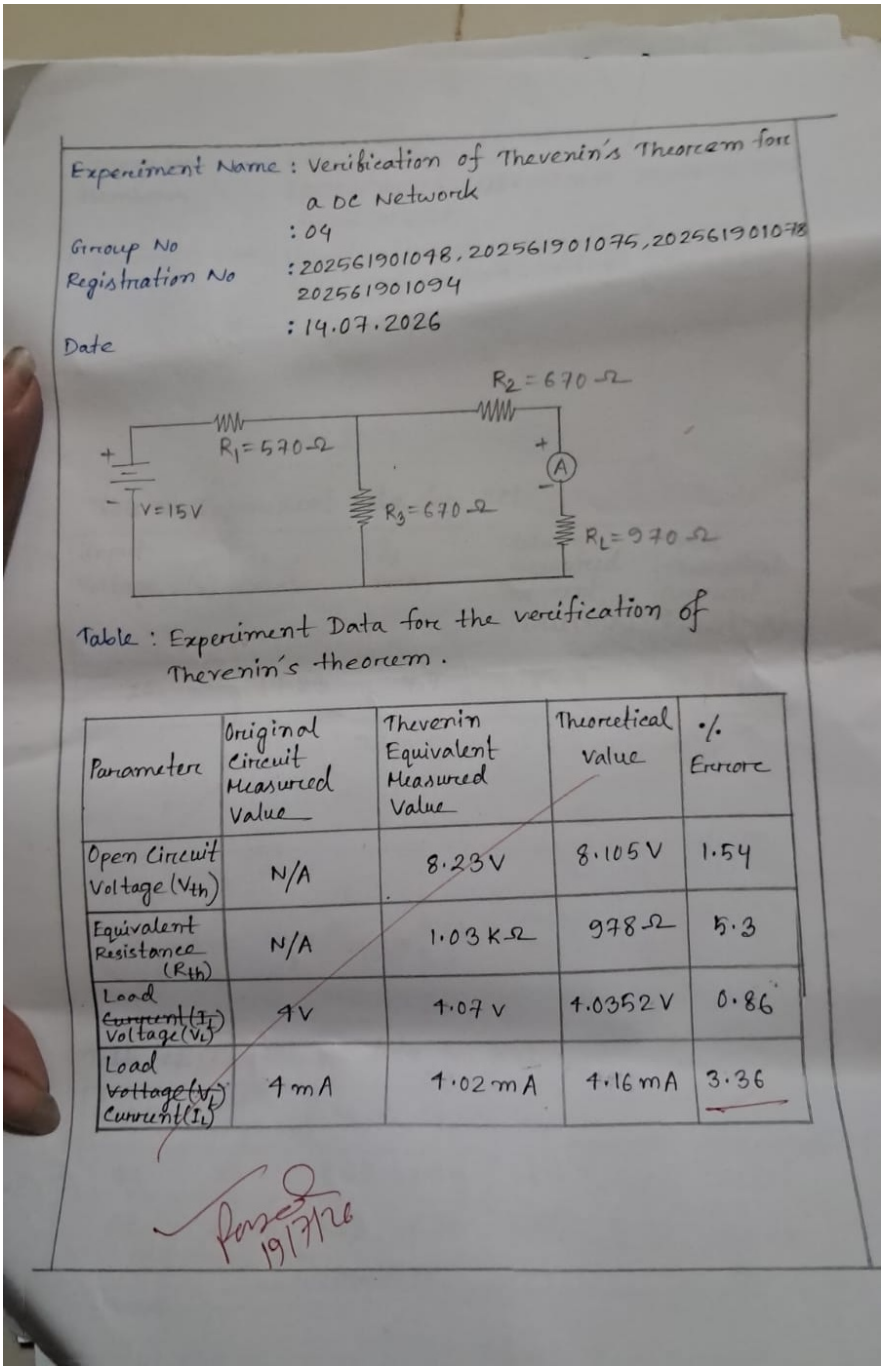
## Conclusion

The experimental results validate Thevenin's Theorem within small operational error limits. The physical equivalent network replicated terminal behavior effectively, proving that a complex linear network can be reduced to a single source with series resistance.

## References


1. Alexander, C. K., & Sadiku, M. N. O. (2016). *Fundamentals of Electric Circuits* (6th ed.). McGraw-Hill Education.
2. Hayt, W. H., Kemmerly, J. E., & Durbin, S. M. (2012). *Engineering Circuit Analysis* (8th ed.). McGraw-Hill Education.
3. Boylestad, R. L. (2015). *Introductory Circuit Analysis* (13th ed.). Pearson.

# Lab Data Sheet



Experiment Name : Verification of Thevenin's Theorem for a DC Network

Group No : 04

Registration No : 202561901098, 202561901075, 202561901078, 202561901094

Date : 14.07.2026




Table : Experiment Data for the verification of Thevenin's theorem.

| Parameter | Original Circuit Measured Value | Thevenin Equivalent Measured Value | Theoretical Value | % Errore |
|---|---|---|---|---|
| Open Circuit Voltage ($V_{th}$) | N/A | 8.23 V | 8.105 V | 1.54 |
| Equivalent Resistance ($R_{th}$) | N/A | 1.03 KΩ | 978 Ω | 5.3 |
| Load ~~Current ($I_L$)~~ Voltage ($V_L$) | 4 V | 1.07 V | 1.0352 V | 0.86 |
| Load ~~Voltage ($V_L$)~~ Current ($I_L$) | 4 mA | 1.02 mA | 4.16 mA | 3.36 |

19/7/26

**Figure 3: Handwritten Lab Data Sheet for Experiment 04**